

14 Singular Value Decomposition



14.2 A Geometric View of SVD

from 2×2 , 3×3 , 3×2 to 2×3 matrices, building intuition for the “rotate $\rightarrow$ scale $\rightarrow$ rotate” steps of singular value decomposition

Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills You Will Master

1

SVD breaks a complex linear transformation into three steps: “rotate $\rightarrow$ scale $\rightarrow$ rotate”

2

Rotation can happen between spaces of different dimensions

3

Scaling can go hand in hand with a change of dimension

4

For a symmetric matrix, SVD is equivalent to spectral (eigen) decomposition

5

The scaling of area or volume in the geometric transformation is set by the singular values

6

A zero singular value means dimension reduction — a vector is projected onto a lower-dimensional space, reflecting a loss of rank

Four matrix shapes, unpacking the three geometric steps of $A=USV^T$

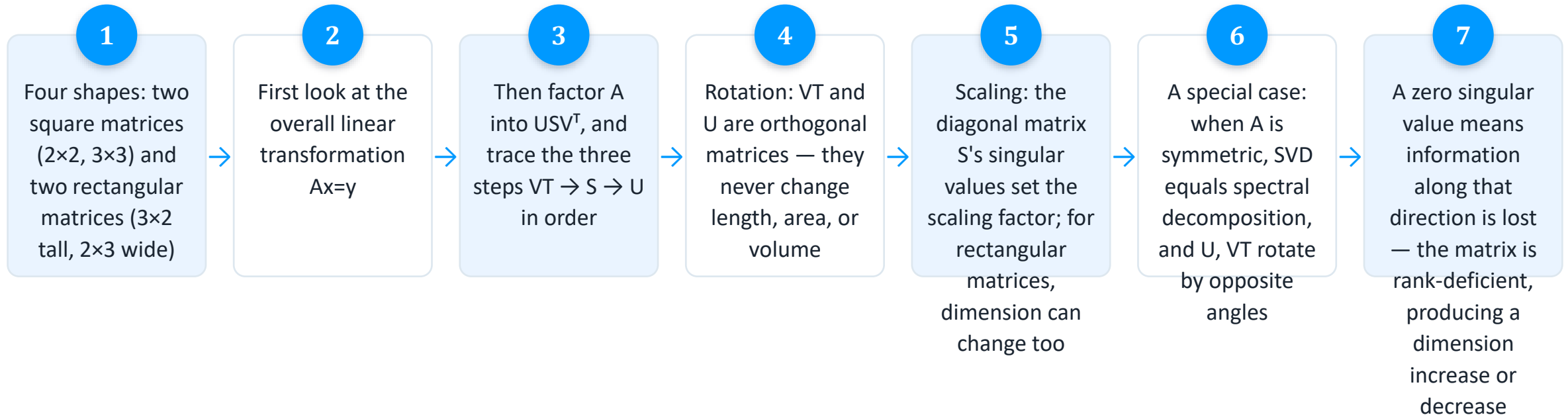


Figure 1 · Linear mappings of matrix A across four different shapes

- This section takes a geometric view of the SVD of four differently shaped matrices: 2×2 , 3×3 , 3×2 , and 2×3
- The shape of A sets the spaces of the input and output vectors: a square matrix keeps the dimension, a rectangular one may raise or lower it

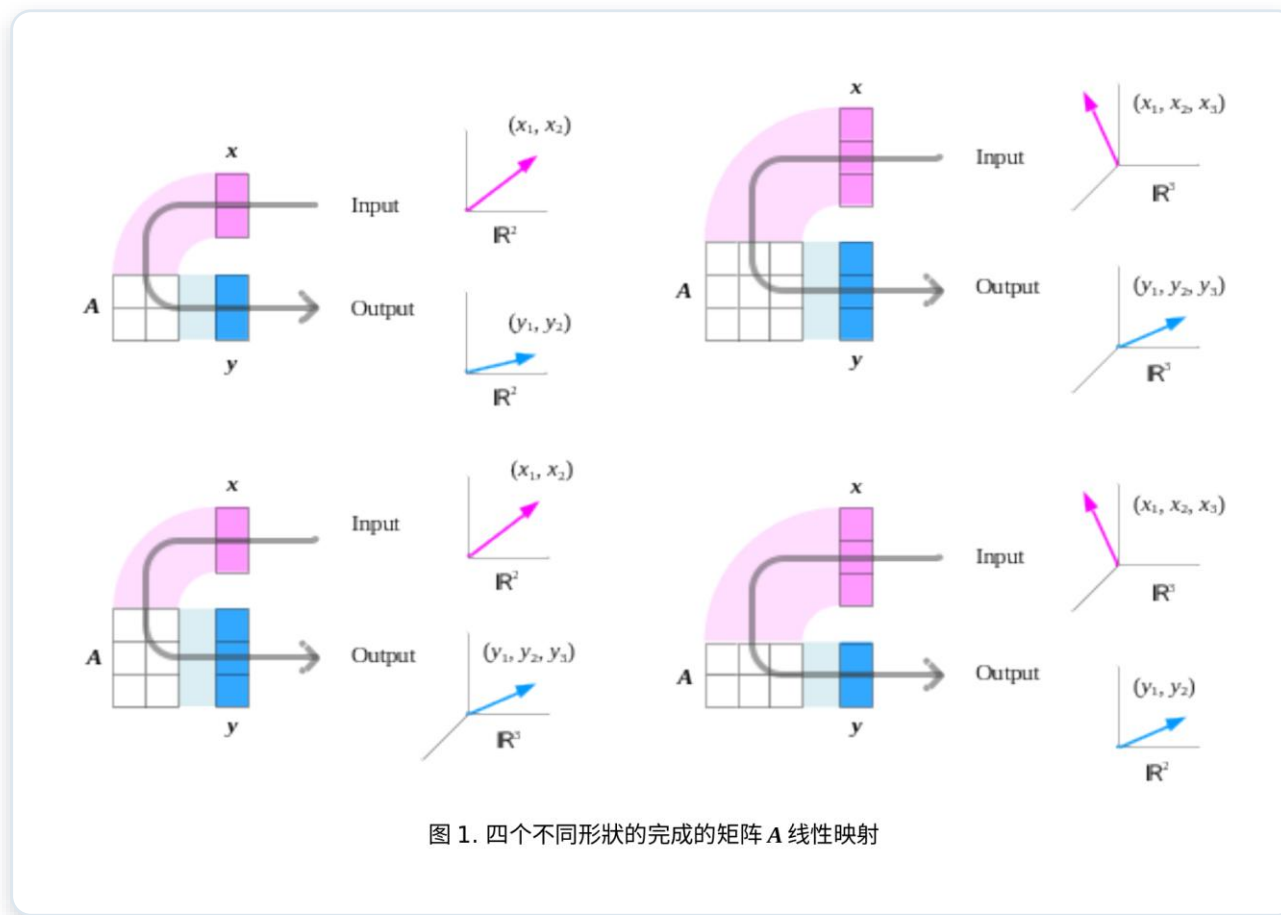


Figure 2 · Linear transformation of a 2×2 symmetric matrix A

- $A = \begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix}$ This geometric transformation is a composite of rotation, shear, and scaling

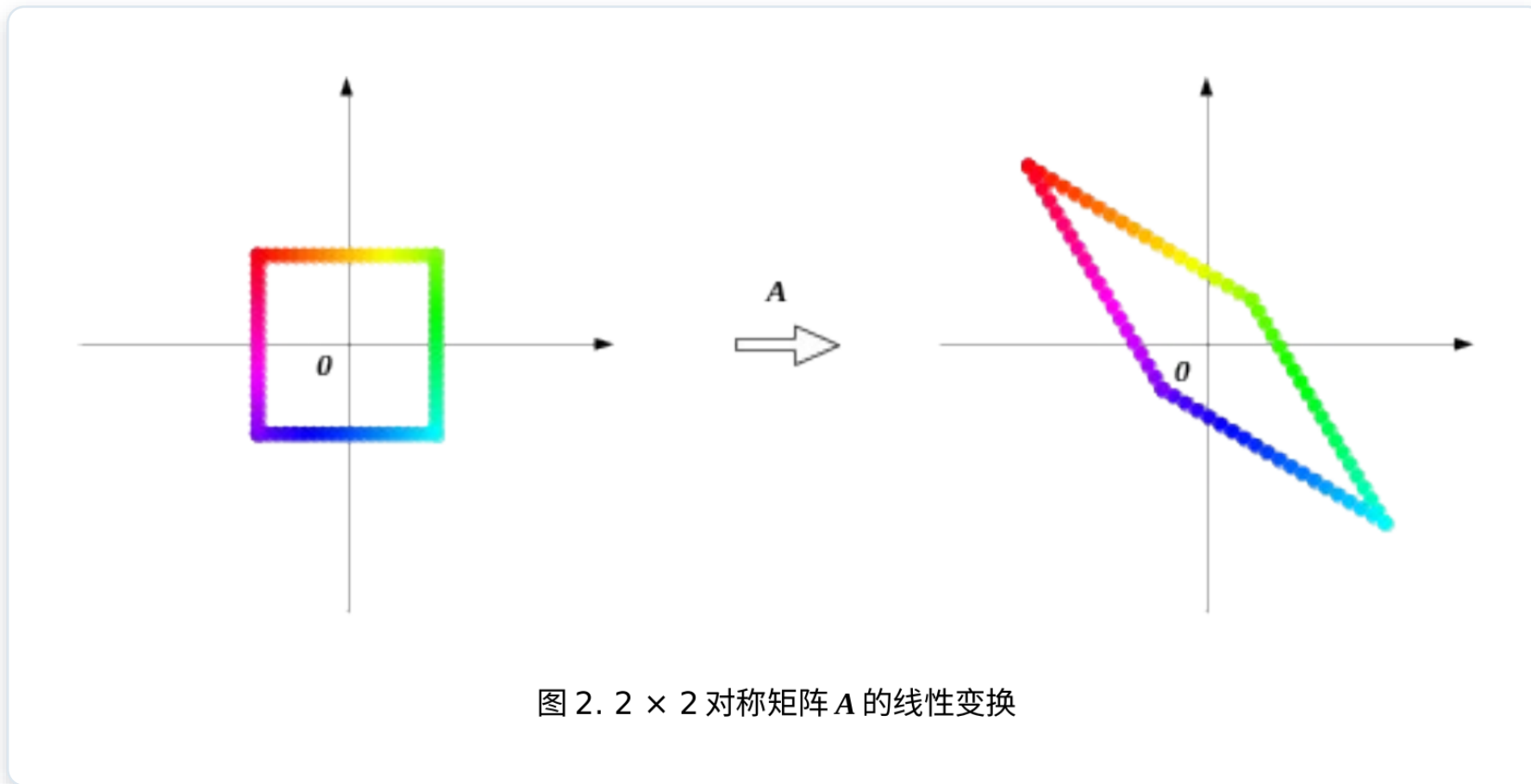


Figure 3 · The transformation of USVT, the SVD of the 2×2 symmetric matrix A

- V^T and U are both rotations that preserve area; $S=\text{diag}(2, 1/2)$ sets the scaling, and $\det(A)=\det(S)=1$
- A is symmetric $\Rightarrow$ its SVD equals its spectral decomposition, and U, V^T rotate by exactly opposite angles

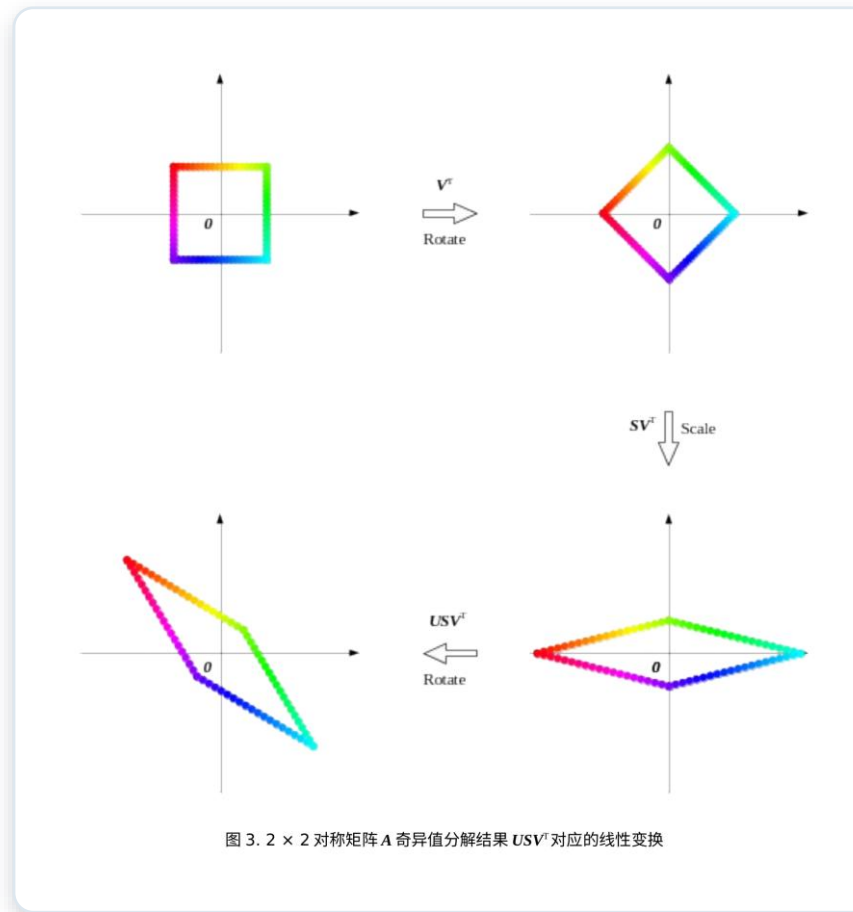


Figure 4 · Linear transformation of a 2×2 non-symmetric matrix A

- $A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$ The geometric transformation of a non-symmetric matrix is likewise a composite of basic operations

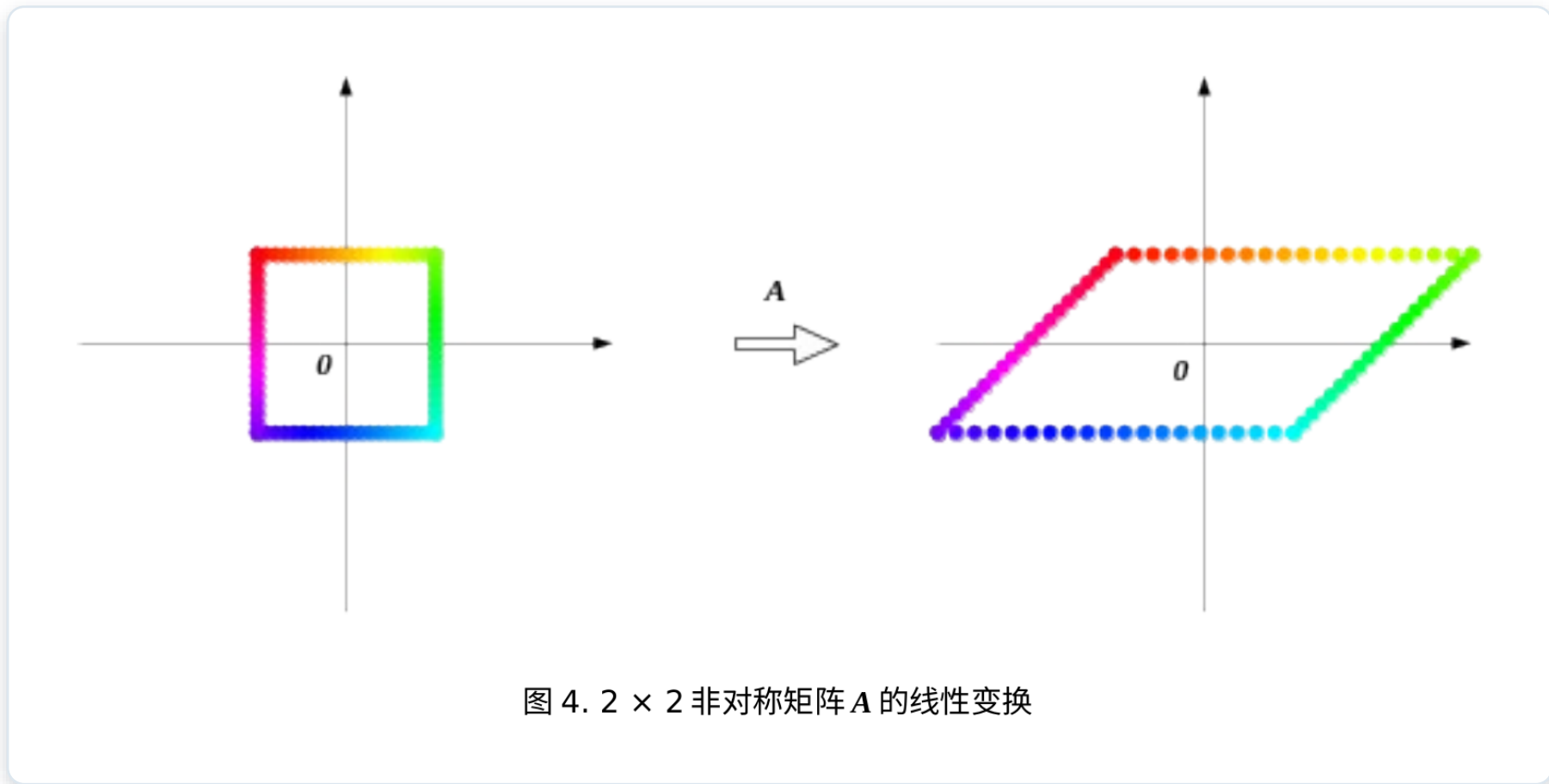


Figure 5 · The transformation of USVT, the SVD of the 2×2 non-symmetric matrix A

- V^T rotates the plane about the origin; $S=\text{diag}(2.288, 0.874)$ stretches along the horizontal axis and shrinks along the vertical axis; U rotates again
- A is not symmetric $\Rightarrow$ the rotation of U is not the inverse of V^T

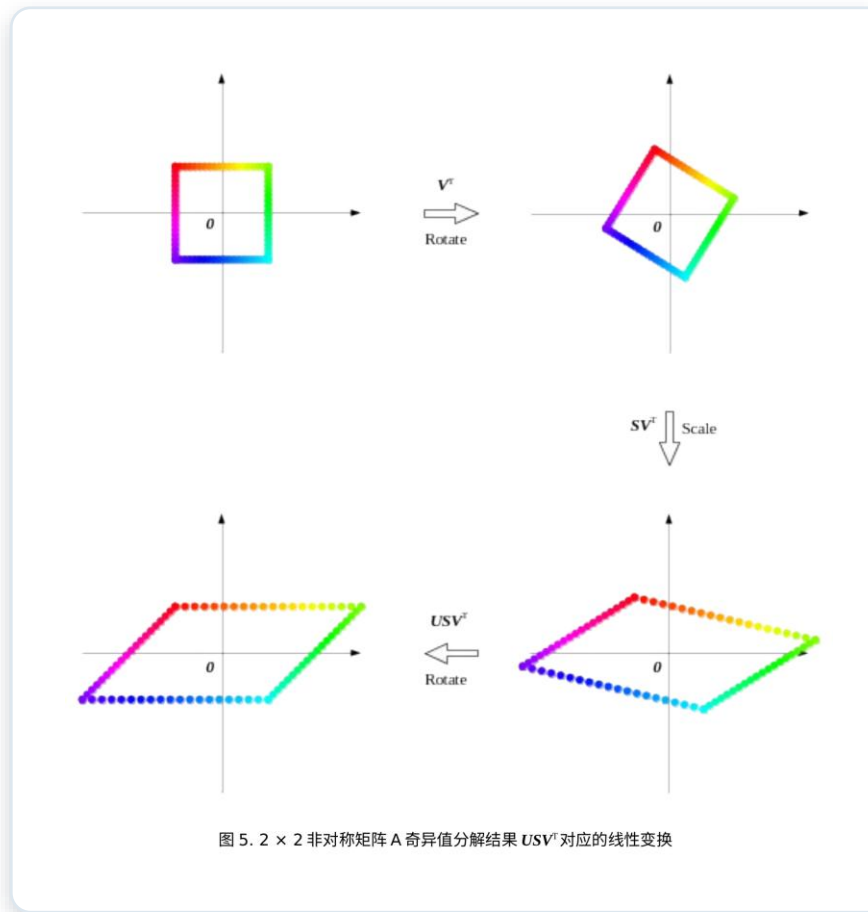


Figure 6 · Linear transformation of a 3×3 matrix A (full rank)

- $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ A full-rank square matrix: its column vectors are not collinear, so its determinant is nonzero

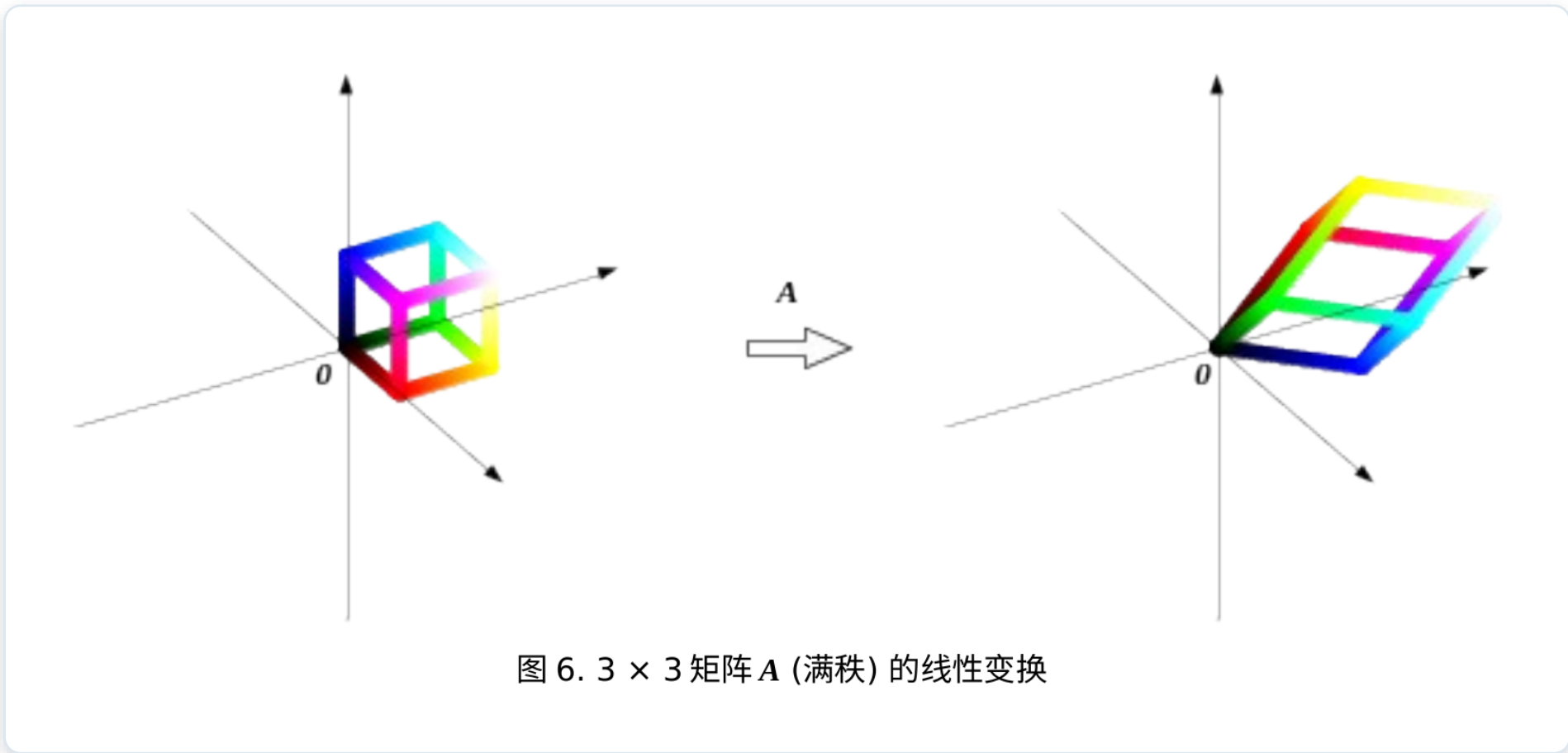


Figure 7 · The transformation of USVT, the SVD of the 3×3 symmetric matrix $\mathbf{A}$ (full rank)

- $\mathbf{V}^T$ and $\mathbf{U}$ are both rotations in 3D space; $\mathbf{S}=\text{diag}(2, 1, 1)$ scales in 3D space
- $\det(\mathbf{S})=2$; since $\mathbf{A}$ is square, $\det(\mathbf{A})=\det(\mathbf{S})=2$

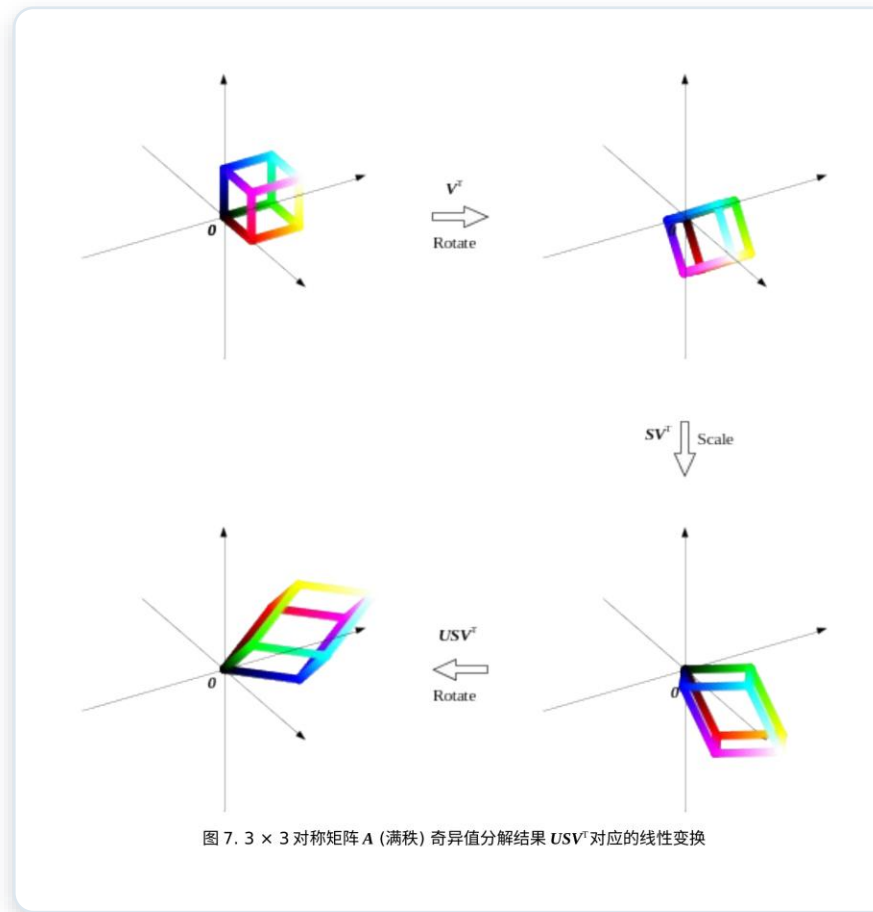


Figure 8 · Linear transformation of a 3×3 matrix A (rank-deficient)

- $A = \begin{bmatrix} 0 & 1 & -1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ $a_1 - a_2 = a_3$, so the columns are linearly dependent, $\det(A)=0$, and the transformation reduces dimension

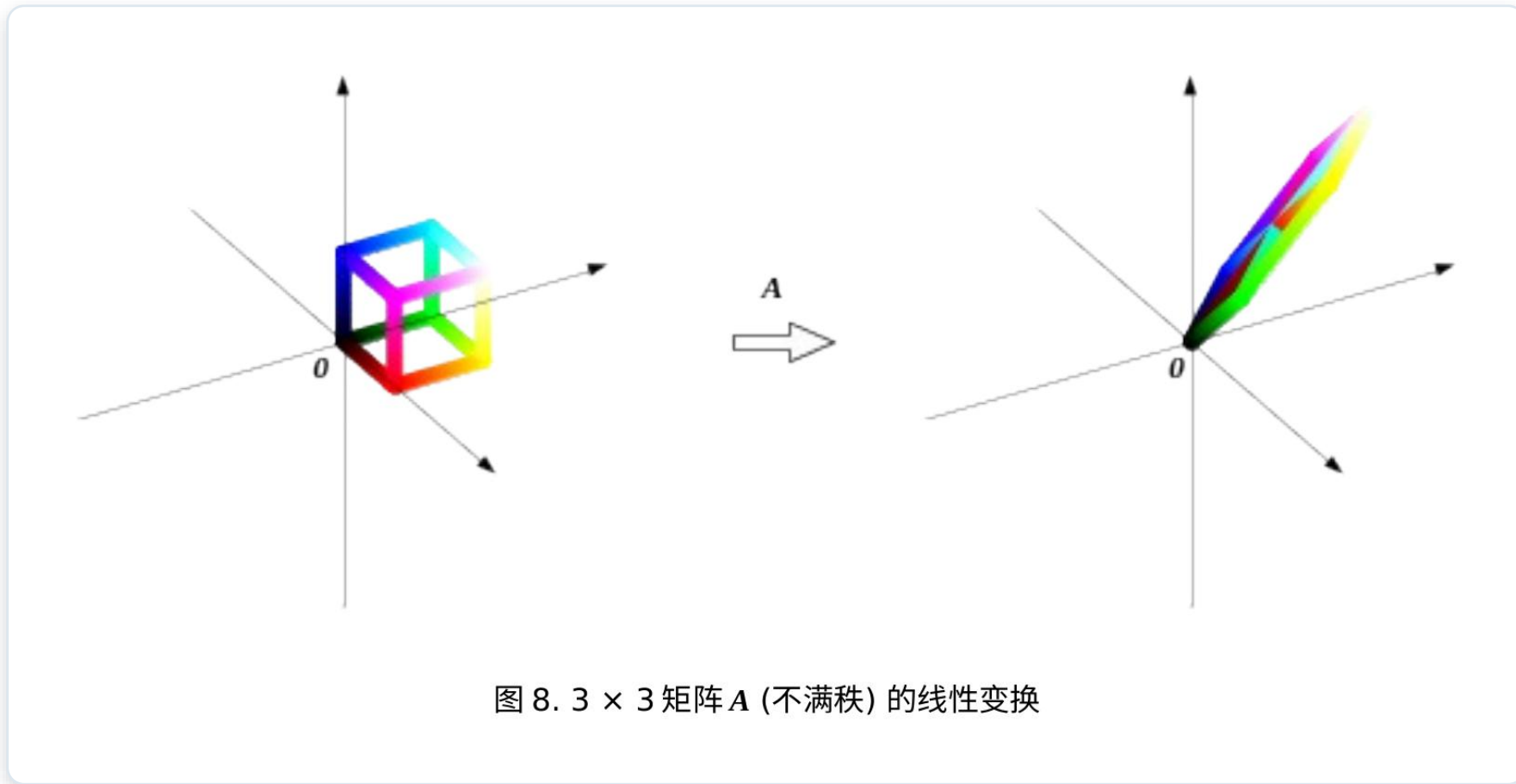
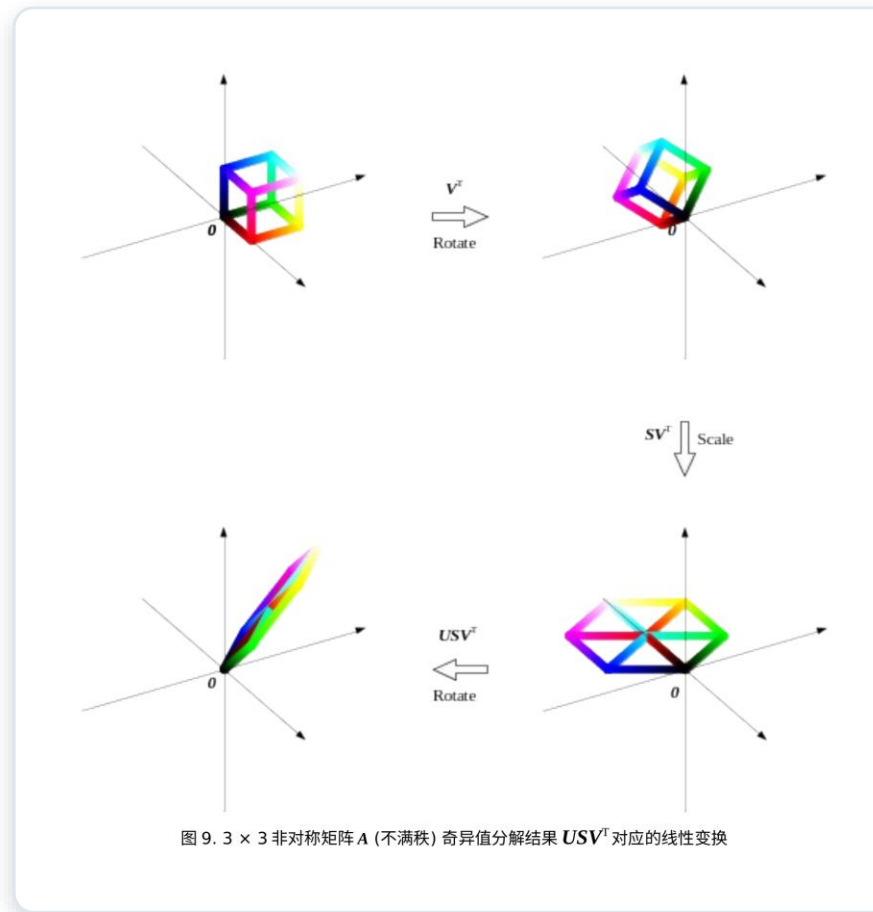


Figure 9 · The transformation of USVT, the SVD of the 3×3 non-symmetric matrix A (rank-deficient)

- V^T and U are both rotations in 3D space; $S = \text{diag}(\sqrt{3}, \sqrt{3}, 0)$, with one singular value equal to 0
- $\text{rank}(A)=2$; U and V are full rank, but S also has rank 2 $\Rightarrow$ scaling comes with a dimension drop



3×2 TALL MATRIX · FULL COLUMN RANK

Figure 10 · Linear transformation of a 3×2 tall matrix A (full column rank)

- $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$ x lives in a plane while $y=Ax$ lives in 3D space, so $Ax=y$ raises the dimension

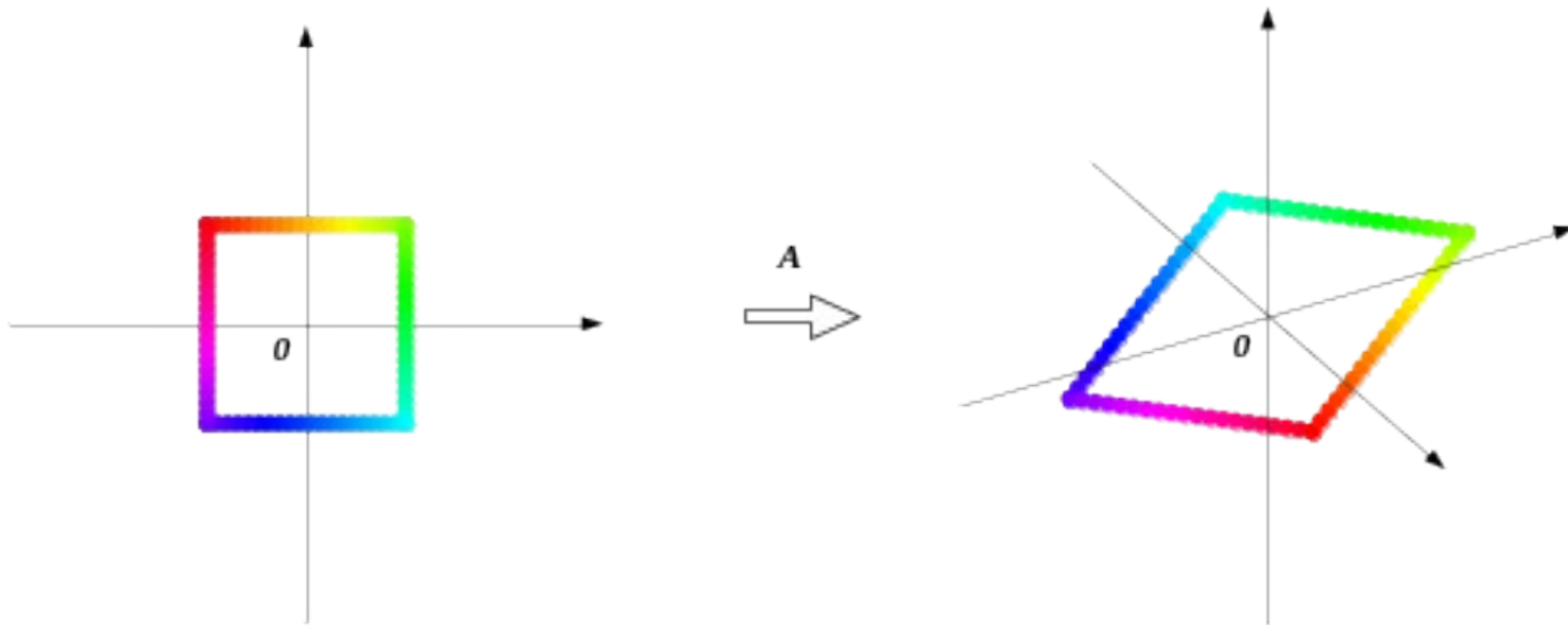


图 10. 3×2 细高矩阵 A (列满秩) 的线性变换

Figure 11 · The transformation of USVT, the SVD of the 3×2 tall matrix A (full column rank)

- V^T rotates the plane about the origin; $S=\text{diag}(\sqrt{3}, 1)$ performs “scale + dimension increase”; U rotates in 3D space
- Like placing a photo (2D) on a table (3D); $\text{rank}(A)=2$, and U, V are full rank

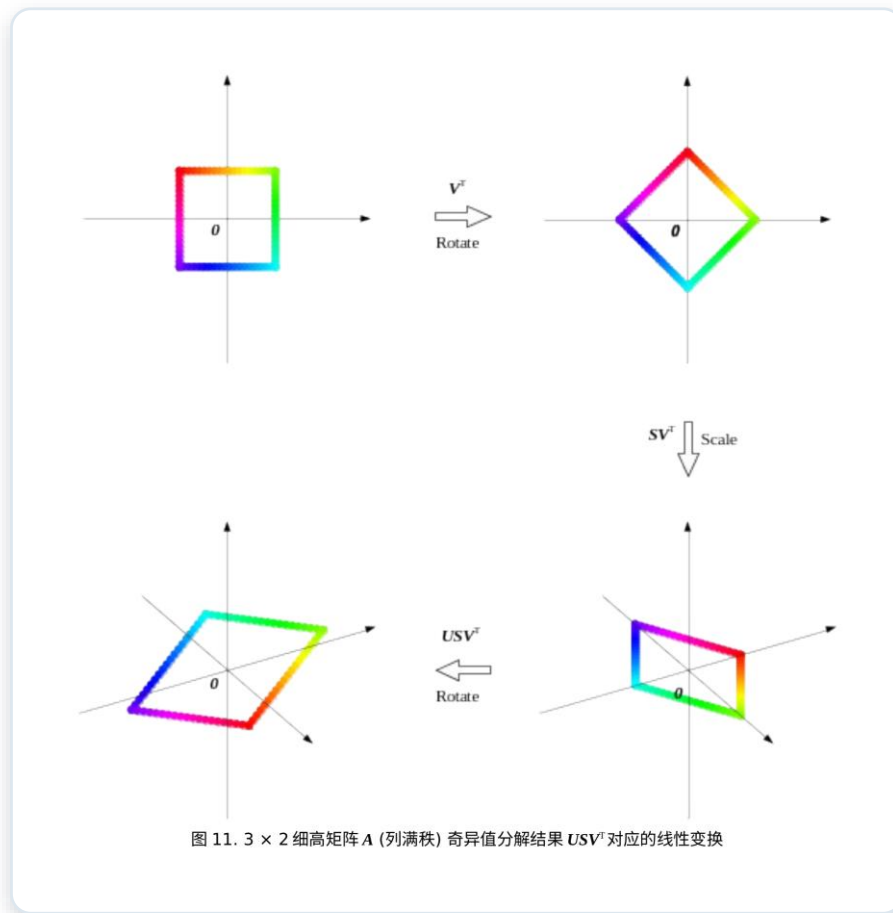


Figure 12 · Linear transformation of a 3×2 tall matrix A (not full column rank)

- $A = \begin{bmatrix} 0 & 0 \\ 1 & -1 \\ 1 & -1 \end{bmatrix}$ The column vectors are collinear, so every transformed vector falls on the same line

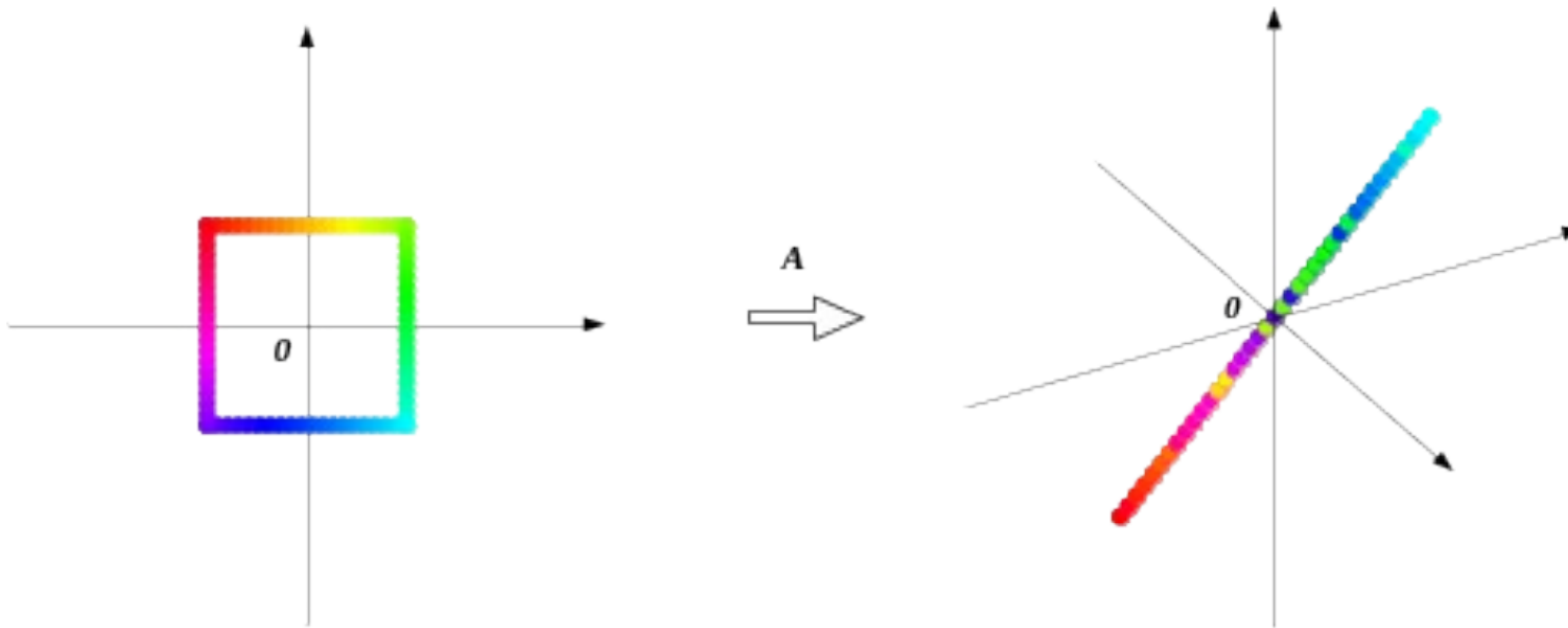
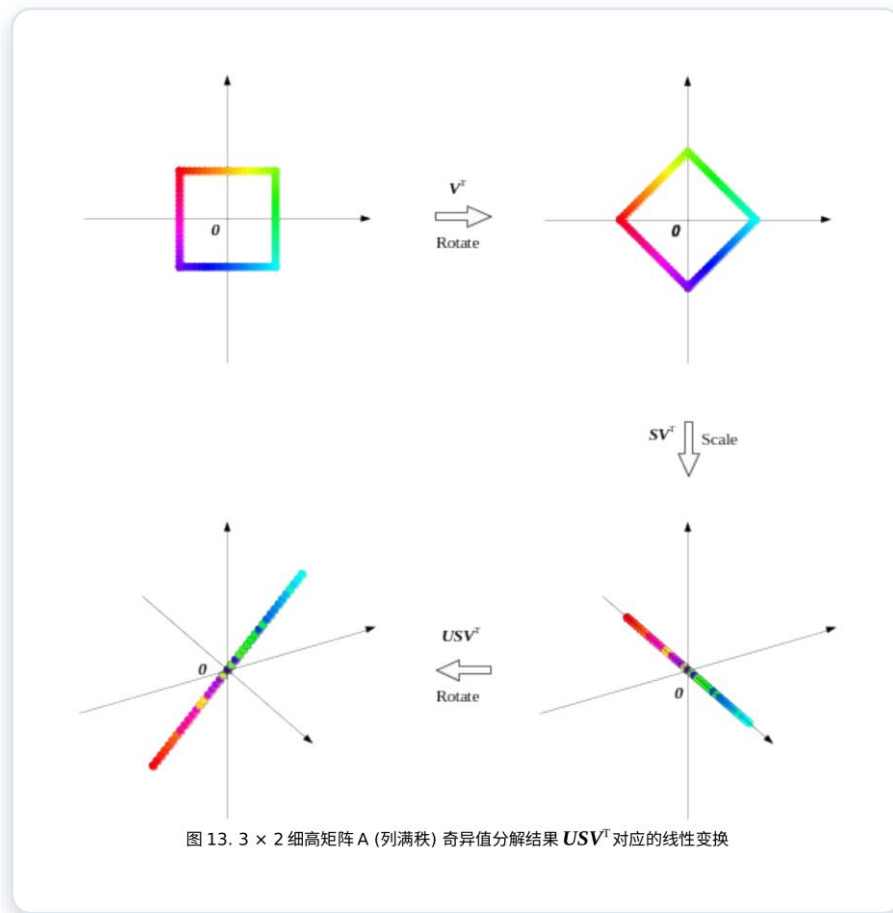


图 12. 3×2 细高矩阵 A (列不满秩) 的线性变换

Figure 13 · The transformation of USVT, the SVD of the 3×2 tall matrix A (not full column rank)

- V^T rotates the plane; one scaling factor of S is 0, which is essentially a dimension drop; U rotates in 3D space
- $\text{rank}(A)=1$; U and V are full rank, but S also has rank 1 $\Rightarrow$ every scattered point ends up on the same line



2×3 WIDE MATRIX · FULL ROW RANK

Figure 14 · Linear transformation of a 2×3 wide matrix A (full row rank)

- $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ x lives in 3D space while $y=Ax$ lives in a plane, so $Ax=y$ reduces the dimension

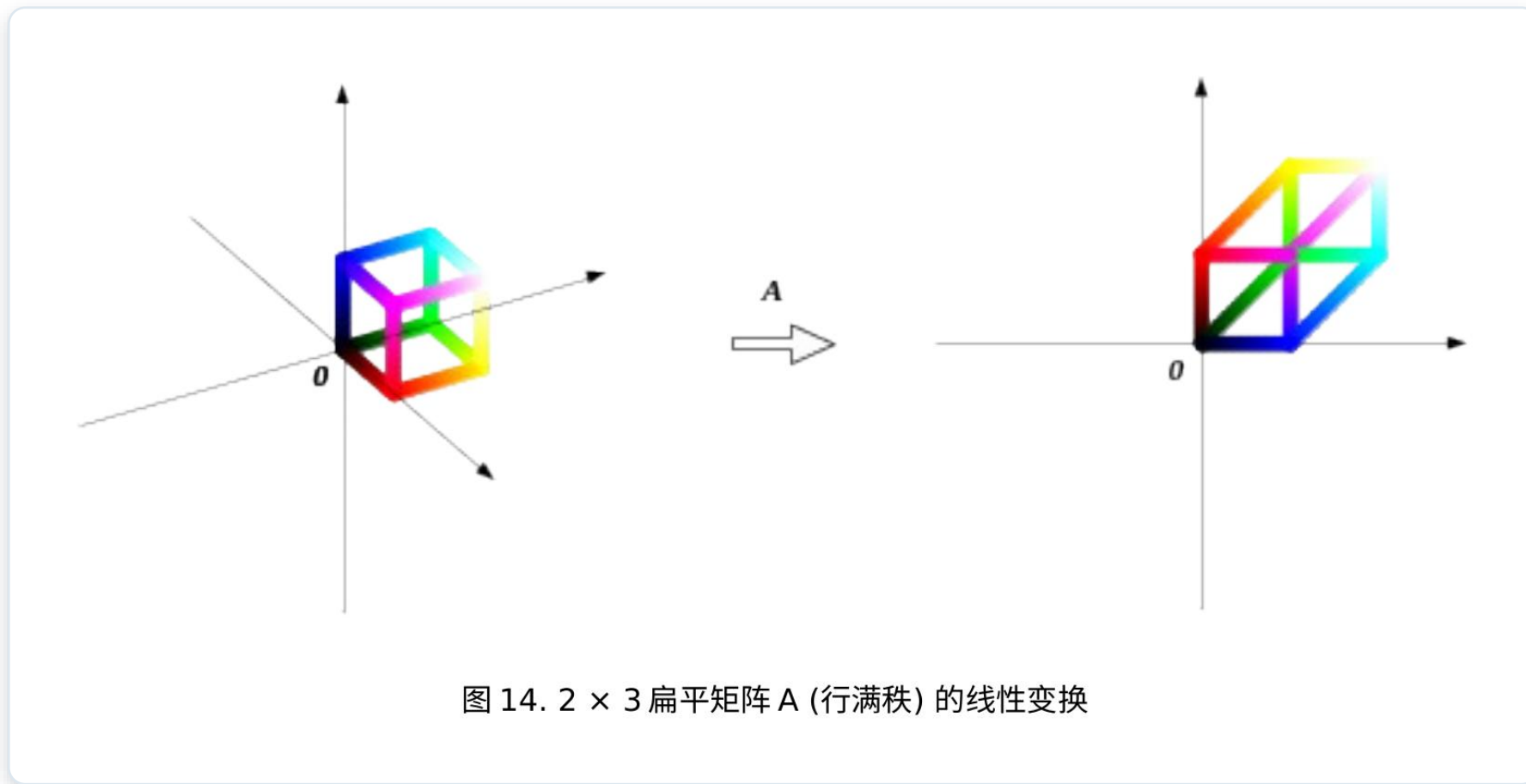


图 14. 2×3 扁平矩阵 A (行满秩) 的线性变换

Figure 15 · The transformation of USVT, the SVD of the 2×3 wide matrix **A** (full row rank)

- V^T first rotates the unit cube in 3D space; $S=\text{diag}(\sqrt{3}, 1)$ performs “scale + dimension reduction”, like taking a photo
- U then rotates the resulting planar shape about the origin; $\text{rank}(\mathbf{A})=2$, and U, V are full rank

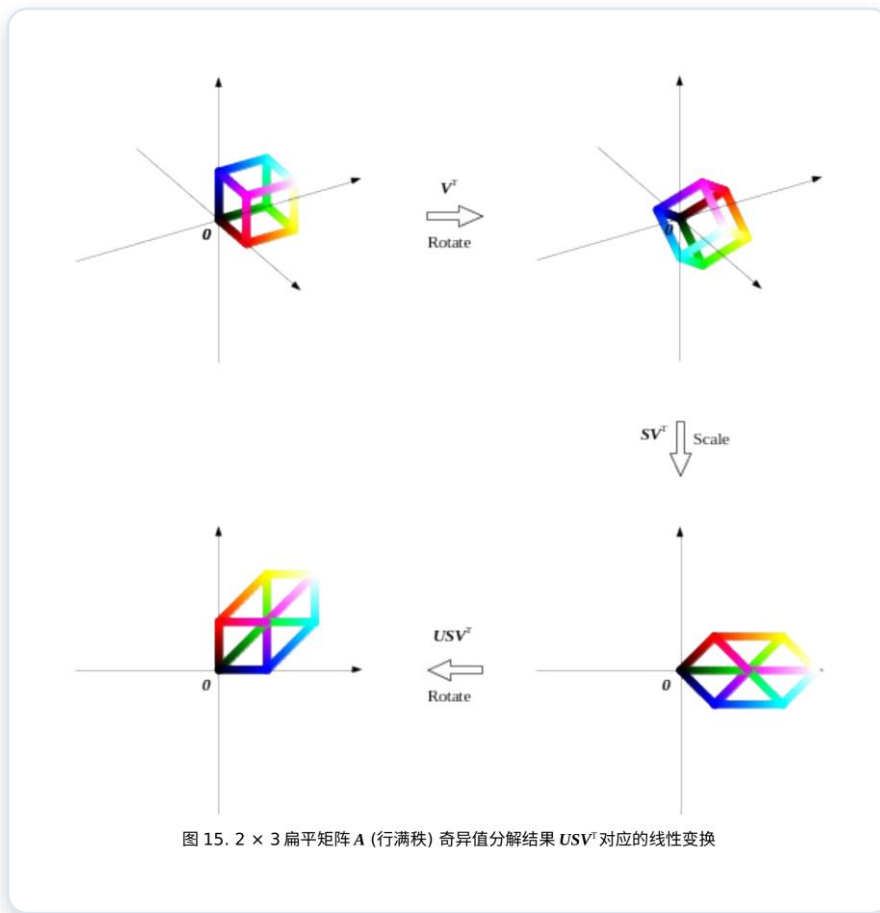


Figure 16 · Linear transformation of a 2×3 wide matrix A (not full row rank)

- $A = \begin{bmatrix} 0 & 1 & 1 \\ 0 & -1 & -1 \end{bmatrix}$ The row vectors are collinear, driving the rank of the result down further

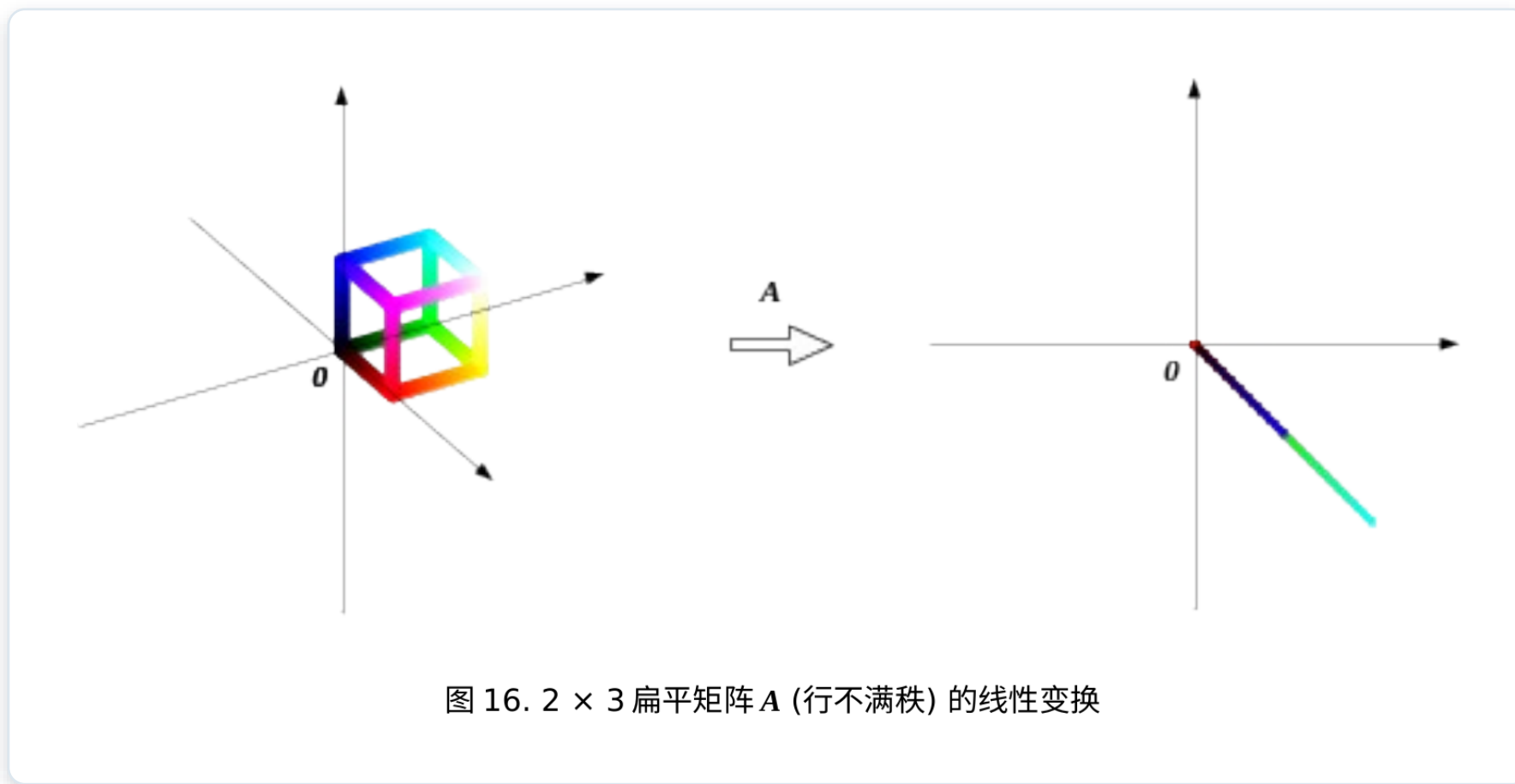
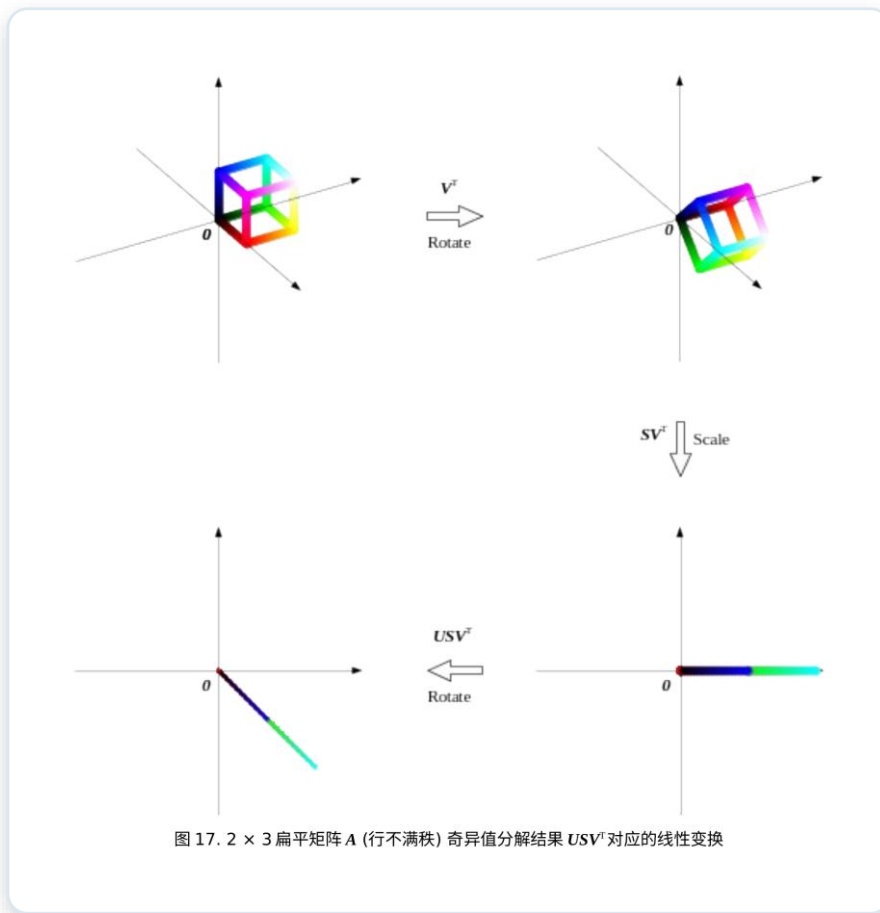


Figure 17 · The transformation of USVT, the SVD of the 2×3 wide matrix **A** (not full row rank)

- $\mathbf{S}=\text{diag}(2, 0)$, with one singular value equal to 0; $\text{rank}(\mathbf{A})=1$, $\mathbf{U}$ and $\mathbf{V}$ are full rank, and $\mathbf{S}$ also has rank 1
- Combine this with the earlier examples and work out for yourself the geometric meaning of the $\mathbf{V}^T$, $\mathbf{S}$, $\mathbf{U}$ steps



SUMMARY

Any matrix A can be broken into three geometric steps: “rotate → scale → rotate”

① $A=USV^T$: orthogonal matrices V^T , U handle rotation; the diagonal matrix S handles scaling

② Rotation never changes length, area, or volume; the scaling factor is set by the singular values

③ When a square matrix is full rank, $\det(A)=\det(S)$ = the product of the singular values

④ When A is symmetric, SVD equals spectral decomposition, and U , V^T rotate by opposite angles

⑤ Rectangular matrices (such as 3×2 or 2×3) raise or lower the dimension while scaling

⑥ A zero singular value $\Rightarrow$ information along that direction is lost, the matrix is rank-deficient, and vectors are compressed into a lower-dimensional space

Key Terms in This Section

Singular Value Decomposition $A=USV^T$ — any matrix can be broken into “rotate $\rightarrow$ scale $\rightarrow$ rotate”

Full Rank / Rank-deficient Whether the column (or row) vectors are linearly independent, deciding whether a singular value is 0

Spectral Decomposition The eigen decomposition $A=V\Lambda V^T$ of a symmetric matrix, equivalent to its SVD

Singular Value A non-negative entry on the diagonal of S ; it sets the scaling factor, and its count bounds the rank

Dimension Increase / Decrease For a non-square matrix, Ax can move the output vector into a space of a different dimension

Orthogonal Matrix The columns of U and V are unit orthogonal vectors; they act as a rotation (or reflection) and preserve length

FURTHER READING

Further Reading

Review · Chapter 2, Section 5

The geometric view of matrix multiplication — the foundation for dimension increase and decrease

Review · Chapter 8, Geometric Transformations

Rotation, scaling, projection, and other basic geometric operations

Review · Spectral Decomposition (EVD)

How a symmetric matrix's SVD relates to its spectral decomposition

Open-source Code Repository

Code and figures for the entire book: <https://github.com/Visualize-ML>

Exercises Q1 – Q9

Q1 By hand, work out the SVD of every example in this section, then check your results in Python with `np.linalg.svd`.

Q3 Plot how the unit square is mapped by the matrix A in Figure 2, overlaying the intermediate results of the VT , S , U steps to reproduce Figure 3.

Q5 Compare U and VT for the symmetric example in Figure 2 and the non-symmetric example in Figure 4, and explain what is special about a symmetric matrix's SVD.

Q7 In Python, verify that $A = \begin{bmatrix} 0 & 1 & -1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ from Figure 8 satisfies $a_1 - a_2 = a_3$, and compute the rank of this matrix.

Q9 Plot the scatter of a unit cube transformed by the matrix A from Figure 8, and check whether the reduced-dimension result degenerates into a plane.

Q2 In Python, verify the SVD of $A = \begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix}$ from Figure 2, and compare it with its spectral (eigen) decomposition.

Q4 Compute the singular values of $A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$ from Figure 4, and verify that $\sigma_1 \times \sigma_2$ equals $|\det(A)|$.

Q6 Use a Laplace expansion to compute the determinant of $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ from Figure 6, and compare it with $\det(S)$.

Q8 Use `np.linalg.matrix_rank()` to check the rank of A in each of the section's 8 examples, and compare it with the number of nonzero singular values in S .

Exercises Q10 – Q17

Q10 Explain why $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$ from Figure 10 raises the dimension, and use Python to verify the transformed result stays within a single plane.

Q12 Compute the singular values $\sqrt{3}$ and 1 of the S matrix in the Figure 10/Figure 11 example, and explain which direction each one scales.

Q14 Compare the full-column-rank example in Figure 10 with the rank-deficient example in Figure 12, and explain what a zero singular value does to the geometric transformation.

Q16 Explain why the example in Figure 14 is called “dimension reduction”, and contrast it with the “dimension increase” example in Figure 10.

Q11 Write a Python function that decides whether an $m \times n$ matrix A raises or lowers dimension, and test it on all 8 examples in this section.

Q13 In Python, verify that $A = \begin{bmatrix} 0 & 0 \\ 1 & -1 \\ 1 & -1 \end{bmatrix}$ from Figure 12 has rank 1, and plot how the transformed points all fall on one line.

Q15 In Python, verify that $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ from Figure 14 compresses the 3D unit cube onto a plane, and plot the resulting hexagonal outline.

Q17 In Python, compute the SVD of $A = \begin{bmatrix} 0 & 1 & 1 \\ 0 & -1 & -1 \end{bmatrix}$ from Figure 16, and verify that it has only one nonzero singular value.

Exercises Q18 – Q25

Q18 Write a Python function `volume_scale(A)` that computes $|\det(A)|$ for a square matrix A , and verify it equals the product of the singular values.

Q20 Explain why the geometric operation of an orthogonal matrix (like U or V in this section) can only be a rotation or reflection, and never changes vector length.

Q22 Drawing on the geometric view of matrix multiplication from Chapter 2, Section 5, explain how the shape ($m \times n$) of a matrix sets the dimensions of the input and output spaces.

Q24 Design a new 3×2 matrix that is not full column rank, work out its SVD by hand as in Figures 12–13, and verify it in Python.

Q19 Pick any 3×3 example from this section and use a sequence of Python scatter plots to show the three intermediate states of VT rotate $\rightarrow$ S scale $\rightarrow$ U rotate.

Q21 In Python, verify that for any example in this section, both $U^T U$ and $V^T V$ equal the identity matrix I .

Q23 In Python, compare the number of singular values, the matrix shape, and the matrix rank across the section's 8 examples, and summarize the general pattern.

Q25 Using Matplotlib or Plotly in Python, pick one example from this section and animate the unit cube's motion through the three SVD steps in 3D space.

10 Prompts to Explore with AI

- 1 Explain why any matrix A can be factored as USV^T , where U and V are orthogonal matrices and S is diagonal.
- 2 Explain why the geometric operation of an orthogonal matrix is a rotation (or reflection), and why it never changes a vector's length.
- 3 Explain what a “singular value” means in singular value decomposition, and how it relates to the rank of a matrix.
- 4 Give an example showing why the SVD of a symmetric matrix is equivalent to its spectral (eigen) decomposition.
- 5 Explain how the shape of a matrix (such as 3×2 or 2×3) determines whether $Ax=y$ raises or lowers dimension.

10 Prompts to Explore with AI

6 Explain why a zero singular value means the information along that direction is lost in the transformation.

7 Using a specific 2×2 matrix as an example, explain the geometric meaning of each of the VT, S, U steps.

8 Explain the relationship between the determinant $\det(A)$ and the singular values, and how it reflects the scaling of area or volume.

9 Compare the scope of eigen decomposition (EVD) and singular value decomposition (SVD), and explain why SVD holds for any matrix.

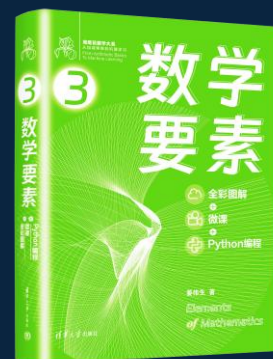
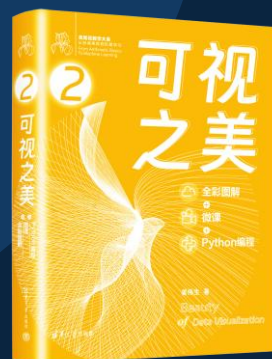
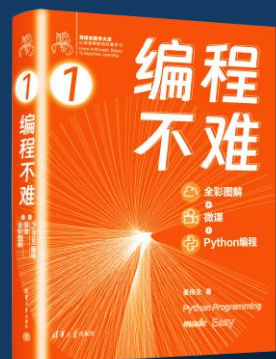
10 Give an example of how singular value decomposition is used in practice for image compression or denoising.

数学

数学基础

线性代数

概率统计



Python编程

数据可视化

工具

数据分析
图和网络

回归、降维
分类、聚类

实践

